

KVK Information System

Krishi Vigyan Kendra database management system. Briefing for explanation and demonstration. October 2026.

This is a Windows desktop application for a KVK office. Staff record farmer visits, on-farm trials, demonstrations, trainings, and extension work. All of that data is stored in one PostgreSQL database named kvk_db. Reports, charts, and the pivot table read the same records.

1. What to say in one minute

The KVK Information System replaces scattered registers with one desktop program. An officer signs in, chooses a module, and saves the activity. The record is written to PostgreSQL immediately. Another officer on the same database sees it. Reports and the pivot table are generated from those saved rows. There is no separate website. The window on the screen is the whole system.

2. Who uses it

Role	What they can do
Admin	Every module, user accounts, and database restore.
Staff	Data entry on most modules. On-farm testing, demonstrations, and extension activities open as view-only. Staff can take a backup but cannot restore.

Sign-in checks the username against the users table. The password is stored as a PBKDF2-SHA256 hash, not as plain text. After five wrong tries, that username is paused for five minutes inside the running program.

3. Modules an officer sees

Module	What is recorded
Visitor Farmers	Farmer, village, district, tehsil, mobile, scientist, and purpose.
On Farm Testing (OFT)	Trial title, crop variety, area, technical assessment t1 t2 t3, and each farmer.
Front Line Demonstrations (FLD)	Demonstration title, crop, area, activity type, and each farmer.
Training Programmes	Title, dates, venue, clientele (PF, RY, EF), and participating farmers.
Vocational Training	Same training fields, used for skill training.
Extension Activities	Kisan mela, group meeting, camp, venue, location, and farmer count.
Other Extension Activities	Radio talk, TV show, newspaper, article, and literature.
Analytics	Counts, charts, and a pivot table.
Reports	Filter by date, department, and module. Export PDF or CSV.
Backup and Restore	Full SQL backup. Restore is for an empty database and is admin-only.
User Management	Create staff or admin accounts, and activate or deactivate them.

Seven departments are seeded: Agronomy, Horticulture, Plant Protection, Veterinary Science, Soil Science, Home Science, and Agricultural Extension. Seasons used by the forms are Kharif and Rabi.

4. How one saved record travels

Step	What happens
1	main.py starts, writes logs/kvk_app.log, and asks the database if it is reachable.
2	database/init_db.py creates any missing tables and the seven departments.
3	The officer signs in. controllers/auth_controller.py checks the users table.
4	The dashboard opens the chosen module.
5	services/activity_service.py finds or creates the farmer, then inserts one activities row.
6	services/audit_service.py writes who did it into audit_logs.

Step	What happens
7	Reports and Analytics read that same row.

5. Technology stack, and why each part is used

The system is a desktop program, not a website. That choice fits an office where people sit at a PC and fill long forms. The pieces below are the ones actually installed and used.

Technology	Version	Why it is here
Python	3.12 on this PC. 3.10 or newer is supported.	The whole program is written in Python: screens, rules, and database calls. One language keeps the office system small enough to run and explain.
PyQt5	5.15	Draws the Windows window: sign-in, forms, tables, calendars, and buttons. It is a desktop toolkit, so the app does not need a browser. It can be packaged as a normal Windows program.
PostgreSQL	13 or newer. This PC uses 18.	The only copy of the office data. It keeps farmers, activities, users, and the audit log after the program is closed. Several PCs can use the same database. Backups are standard SQL files.
psycopg2	2.9	The driver that lets Python speak to PostgreSQL. database/session.py uses it.
SQLAlchemy	2.0	Turns database tables into Python objects in models/entities.py. Services ask for farmers and activities without opening a new connection style in every screen.
ReportLab	4.2	Builds the PDF reports, including department-wise columns.
passlib	1.7	Hashes passwords with PBKDF2-SHA256 before they are stored.
python-dotenv	1.0	Reads .env so the database address stays out of the program code.
PyInstaller	6.12	Can pack the program into KVKSystem.exe for a PC that should start without a programmer.

6. Why PostgreSQL, not a spreadsheet

A spreadsheet is easy to start and easy to lose. Two people editing it create two copies. This office needs one shared record of farmer work, with dates, departments, and an audit of who changed what.

PostgreSQL is a database server. The KVK program does not store the records inside the program file. It sends them to PostgreSQL. If the program is closed, the data remains. pg_dump can copy the whole database to a .sql file. Indexes on activity date, department, module, and farmer keep filters and reports usable as the register grows.

A second office can use the same data. The intended remote setup is an SSH tunnel: the far PC still connects to localhost, and the tunnel carries the encrypted SQL to the host machine. The program itself does not contain a web server.

7. Why PyQt5, not a website

The daily work is data entry: dates, districts, tehsils, farmer batches, and long forms. PyQt5 gives a native Windows window with combo boxes, calendars, and tables. Officers do not open a browser or depend on a web host. The same screens run on the database host and, through the tunnel, on a remote PC. The green office look comes from ui/styles.py.

8. Which file connects the database

No screen connects by itself. The chain is:

Order	File	Job
1	.env	Holds DB_HOST, DB_PORT, DB_NAME, DB_USER, and DB_PASSWORD. The password is not printed in this document.
2	database/config.py	Reads .env and builds the PostgreSQL address. Next to the EXE, it reads the .env beside the EXE.
3	database/session.py	Opens the connection. Every other file uses this. The pool keeps 10 connections ready.
4	database/init_db.py	On startup, checks the connection, creates missing tables, and seeds departments.
5	main.py	Runs that check before sign-in.
6	services	Activity, report, analytics, user, backup, and audit services all open a session from session.py.

The live database name is kvk_db. The application name sent to PostgreSQL is KVKSystem.

9. The six tables

Table	Holds
departments	The seven department names.
users	Username, roles (admin or staff), active flag, and password hashes.
farmers	Farmer name, village, and 10-digit mobile. The same person is matched on those three values.
activities	Every visitor, OFT, FLD, training, and extension row. Module-specific fields are columns on this table.
reports	A log of exported report files and the filters used.
audit_logs	Create, edit, delete, backup, restore, and user changes.

models/entities.py is the definition the program uses. database/schema.sql is a written copy of that design. Startup creates tables from the Python models, then adds any newer columns that an older database does not have yet.

10. Source files

File	Used for
main.py	Start, logging, database check, sign-in.
controllers/auth_controller.py	First admin account and login check.
ui/login_window.py	Full-screen sign-in.
ui/dashboard_window.py	Top navigation and module pages.
ui/activity_form_widget.py	Entry forms and lists for the seven activity modules.
ui/reports_widget.py	Report filters, table, PDF, and CSV.
ui/backup_widget.py	Backup and restore.
ui/user_management_widget.py	User accounts.
ui/styles.py	Colors, buttons, and tables.
analytics/analytics_widget.py	Dashboard charts and pivot screen.
analytics/analytics_service.py	Count, chart, and pivot queries.
services/activity_service.py	Save, edit, delete, and list activities.
services/report_service.py	Chooses report columns and builds export rows.
services/user_service.py	Create and activate users.
services/backup_service.py	pg_dump and psql restore.
services/audit_service.py	Audit log rows.
reports/pdf_builder.py	ReportLab PDF layout.
ui/oft_flow_widget.py	An older OFT screen. The dashboard does not open it.

11. Reports follow the department

Open Reports, choose the department and the module, then Apply filters. The table, the PDF, and the CSV use the same columns. The PDF title includes the department, for example KVK Activity Report - Agronomy - On Farm Testing (OFT).

Choice	Columns the report keeps
Agronomy + On Farm Testing	Date, title, crop, farmer count, farmer, village, mobile, scientist, purpose, district, tehsil, t1, t2, t3, area, activity, and season.
Any other department + On Farm Testing	The same trial fields, without season. Season is collected only for Agronomy.
Horticulture + Demonstrations	Demonstration fields, including season.
Any other department + Demonstrations	Demonstration fields, without season. Season is collected only for Horticulture.
Any department + Training	Training title, dates, type, venue, clientele, farmer count, and each farmer. Season is not a training field.
Extension	Date, activity, venue, location, purpose, and number of farmers.
Other extension	Date, activity, and title of the show, talk, or article.

The default date range is the last one year, so the demonstration records are included. Change From and To when a shorter period is needed.

12. Analytics and the pivot

Analytics reads the activities table inside PostgreSQL. The dashboard shows activity counts, participants, department charts, and a time trend. Pivot Analysis lets you pick rows, columns, and a value, then Generate Pivot. Double-click a pivot row to see the records behind it. With the demonstration data loaded, a department-by-season count returns all seven departments.

13. Demonstration data already loaded

Forty-seven Maharashtra records are in kvk_db for the presentation. They cover Pune, Nashik, Ahmednagar, Solapur, Jalgaon, Satara, Kolhapur, Amravati, Aurangabad, and Nagpur.

Module	What you can open
Visitor Farmers	12 farmers, including Baramati, Niphad, Rahuri, and Pandharpur.
On Farm Testing	Soybean JS 20-98 in Pune, and rabi onion Phule Samarth in Nashik.
Demonstrations	Bt cotton in Jalgaon, and pomegranate Bhagwa in Solapur.
Training	Soybean sowing, a soil health card camp, and a dairy first-aid vocational course.
Extension	Kisan mela, group meeting, diagnostic visit, animal health camp, mahila melava, and others.
Other extension	Radio talk, TV programme, newspaper coverage, article, and a literature folder.

14. Backup, log, and safety

Backup uses pg_dump and writes a .sql file. Restore is blocked when farmers or activities already exist, so a restore cannot silently duplicate the register. A safety copy can be written first. The PostgreSQL tools are found from PG_BIN_DIR in .env or from a normal PostgreSQL install.

logs/kvk_app.log keeps startup and crash messages. It rotates after about 1 MB and keeps five older files. Unexpected errors show a short message on screen and the detail goes into that log.

15. How to start it for the demonstration

PostgreSQL must be running. Then, in PowerShell:

```
cd C:\Users\ravir\Downloads\KVK-main\KVK-main
py -3.12 main.py
```

Sign in with the admin account already stored in the database. The username is admin. The password is the ADMIN_PASSWORD value in .env. Do not read the database password aloud. That password only connects the program to PostgreSQL. It is not the officer sign-in password.

A built EXE also exists at dist\KVKSystem.exe, with a copy of .env beside it. For the explanation, running main.py is enough, because the code and the database are the same ones this document describes.

16. Short answers if someone asks

Question	Answer
Where is the data?	In PostgreSQL, database kvk_db, mainly in the activities table.
Is it a website?	No. It is a PyQt5 desktop program.
Can two PCs use it?	Yes, if both connect to the same PostgreSQL server. Remote PCs are intended to use an SSH tunnel.
Are passwords visible in the database?	No. They are PBKDF2-SHA256 hashes.
What if the PC is switched off?	The database server still holds the data. The program is only the window.
How is a report different for each department?	The selected department decides which fields are columns. Agronomy trials include season and t1 t2 t3. Horticulture demonstrations include season. Training never uses season.
Who can restore the database?	Only an admin, and only when the database is empty.